

Theena's Engineering Career Launch Program

An 8-week program of research and essay assignments for an 18-year-old exploring mechanical or electrical engineering, designed to convert curiosity into a concrete plan: an entry-level job at an engineering firm, tuition reimbursement, and a degree path — instead of dead-end jobs.

What's in this folder

File	What it is
assignments/week-01 ... week-08	One assignment per week: a research task + an essay. Hand out one at a time.
resources/entry-level-jobs.md	The actual job titles to search for (most people don't know these exist)
resources/tuition-reimbursement-companies.md	Employers known for paying for school, and how to verify current programs
resources/education-pathways.md	Community college → transfer, apprenticeships, and why ABET matters
resources/internships-and-coops.md	How internships and co-ops actually work, what's open before college enrollment, and the self-made internship pitch
resources/north-texas-starter-map.md	Researched map of McKinney/Collin County employers, colleges, and apprenticeships (verified Aug 2026)
resources/essay-rubric.md	How to evaluate each essay so feedback is fair and consistent
tracker.md	A simple checklist to track progress and deliverables

The design behind the program

Each week produces a **real artifact she keeps** — a job list, a resume, a budget, an application actually submitted. The essays aren't busywork; each one forces her to process what she found and commit to a position. By week 8 she has:

1. A researched picture of what mechanical and electrical engineers (and technicians) actually do all day
2. A list of 15+ real, local, entry-level openings at engineering firms and manufacturers
3. A resume and cover letter targeted at those roles
4. A financial comparison: dead-end job vs. technician job + tuition reimbursement, in dollars
5. At least 2 informational interviews with people doing the work
6. At least 3 submitted applications

7. A written 5-year plan she authored herself

How to use this (a note for you, the parent)

- **One assignment per week.** Don't hand her the whole stack — it reads as a wall of homework instead of a project.
- **Let her present, not just submit.** The best version of this is a weekly 20-minute conversation where she walks you through what she found. The essay is her prep for that conversation.
- **Tie effort to real stakes, not just screen time.** The program works best when finishing an assignment unlocks something she wants, and when *she* starts seeing the payoff (Week 4's money math usually does this — the dollar gap between a dead-end job and a technician-plus-tuition path is genuinely startling).
- **Expect the goal to shift, and let it.** If her research convinces her that, say, an electrician apprenticeship or an engineering technology degree fits her better than a BS in engineering, that is the program *working* — she's making an informed decision instead of drifting. The win condition is “a researched plan she owns,” not any one specific plan.
- **She should own the artifacts.** The job list, resume, and 5-year plan are hers. Your role by Week 8 is editor and cheerleader, not author.

Weekly overview

Week	Theme	Research deliverable	Essay
1	What do engineers actually do?	Comparison table: ME vs. EE, day-to-day reality	“Which of these lives do I actually want, and why?”
2	The hidden front door: technician & entry-level roles	List of 10 real job titles + what they pay locally	“The jobs nobody told me about”
3	Mapping the local market	15 real openings at real companies within commuting range	“Three companies I could see myself in”
4	The money: dead-end job vs. career path	Spreadsheet: 5-year earnings + tuition reimbursement math	“What my time is actually worth”
5	Education pathways	Comparison of community college, apprenticeship, and 4-year routes with real local costs	“My education plan and why I chose it”
6	Becoming a candidate	Resume + cover letter targeted at Week 3 openings	Cover letter (the essay IS the deliverable)
7	Talking to real people	2+ informational interviews completed, with notes	“What the people doing this job told me”
8	The plan	3+ applications submitted	“My five-year plan” (capstone)

Ground rules for every assignment

Written for Theena — these appear on each assignment sheet too:

1. **Primary sources beat vibes.** Cite where every fact came from (job posting, BLS page, a person you talked to). “I read somewhere” doesn’t count.
2. **Real numbers, your area.** Salaries and tuition costs must be for your metro area, not national averages, wherever possible.
3. **Essays are 600-900 words** unless the assignment says otherwise. Long enough to think, short enough to finish.
4. **AI is a research assistant, not a ghostwriter.** Using AI to find sources or explain a concept is fine. The essay must be in your own words — the weekly conversation will make it obvious if it isn’t.